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MACHINE LEARNING (25SC2107E)

Y25 - 2026-2027 Trimester - 4

PROJECT ABSTRACT SUBMISSION FORM

SEC NO. 05

Wine Type Classification (Red/White) Using Regularized Logistic Regression

1. ABSTRACT

Abstract

The project “Wine Type Classification (Red/White) Using Regularized Logistic Regression” is a machine learning-based system developed to classify wine samples into Red Wine or White Wine based on their chemical properties. The system uses physicochemical features such as fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates, and alcohol. The collected wine dataset is processed and prepared using Python, Pandas, and NumPy. Google Colab is used as the development environment for data preprocessing, model training, testing, and evaluation. Relevant features are selected and scaled before being provided as input to a Regularized Logistic Regression model using Scikit-learn, which predicts whether a wine sample belongs to the Red or White category. The model is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix to measure its classification performance.

The system is designed for wine analysis, students, researchers, and users who want to understand the relationship between wine chemical properties and wine type. L1 (Lasso) and L2 (Ridge) regularization are considered to control model complexity, reduce the effect of irrelevant features, and help prevent overfitting. The trained machine learning model can be further connected to a Flask backend/API, allowing users to provide wine chemical measurements and receive a classification result. MySQL can be used to store wine data, user information, and prediction records for future analysis and model improvement.

ABSTRACT — CONTINUATION

The project also provides a basis for comparing the effect of different regularization approaches on classification performance. Proper preprocessing, feature scaling, and selection of the regularization parameter are used to obtain a reliable model. Security and input validation can be incorporated when the model is deployed as an application. By combining wine chemical data, data preprocessing, Google Colab, Regularized Logistic Regression, and model evaluation, the system provides an easy and data-driven way to classify wine samples as Red or White and demonstrates the practical use of machine learning for binary classification.

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